

Use the midpoint rule to find a formula to approximate the following integral using n subintervals.

$$\int_{-3}^2 x^2 + 6x + 7 \, dx, \quad \Delta x = \frac{2 - (-3)}{n} = \frac{5}{n}, \quad x_k = -3 + \left(k - \frac{1}{2}\right) \frac{5}{n}$$

$$A_n = \Delta x \sum_{k=1}^n f(x_k) = \frac{5}{n} \sum_{k=1}^n \left[-3 + \left(k - \frac{1}{2}\right) \frac{5}{n} \right]^2 + 6 \left[-3 + \left(k - \frac{1}{2}\right) \frac{5}{n} \right] + 7$$

use WolframAlpha : expand $(-3 + (k - 1/2) * 5/n)^2 + 6(-3 + (k - 1/2) * 5/n) + 7$

$$= \frac{5}{n} \sum_{k=1}^n \left[\frac{25}{n^2} k^2 - \frac{25}{n^2} k + \frac{25}{4n^2} - 2 \right]$$

$$= \frac{5}{n} \left[\frac{25}{n^2} \frac{n(n+1)(2n+1)}{6} - \frac{25}{n^2} \frac{n(n+1)}{2} + \frac{25}{4n^2} n - 2n \right]$$

$$\text{simplify } (25/n^2)n(n+1)(2n+1)/6 - (25/n^2)n(n+1)/2 + (25/(4n^2))n - 2n$$

$$= \frac{5}{n} \left(\frac{76n^2 - 25}{12n} \right)$$

$$\boxed{A_n = \frac{380n^2 - 125}{12n^2}}$$

Use the trapezoidal rule to find a formula to approximate the following integral using n subintervals.

$$\int_{-3}^2 x^2 + 6x + 7 \, dx, \quad \Delta x = \frac{2 - (-3)}{n} = \frac{5}{n}, \quad x_k = -3 + k \frac{5}{n}$$

$$x_{k-1} = -3 + (k-1) \frac{5}{n}$$

$$A_n = \Delta x \sum_{k=1}^n \frac{1}{2} [f(x_k) + f(x_{k-1})] = \frac{5}{n} \frac{1}{2} \sum_{k=1}^n \left[(-3 + k \frac{5}{n})^2 + 6(-3 + k \frac{5}{n}) + 7 \right] +$$

$$\left[(-3 + (k-1) \frac{5}{n})^2 + 6(-3 + (k-1) \frac{5}{n}) + 7 \right]$$

use WolframAlpha expand $((-3 + k \cdot 5/n)^2 + 6(-3 + k \cdot 5/n) + 7) + ((-3 + (k-1) \cdot 5/n)^2 + 6(-3 + (k-1) \cdot 5/n) + 7)$

$$= \frac{5}{2n} \sum_{k=1}^n \left(\frac{50}{n^2} k^2 - \frac{50}{n^2} k + \frac{25}{n^2} - 4 \right)$$

$$= \frac{5}{2n} \left[\frac{50}{n^2} \frac{n(n+1)(2n+1)}{6} - \frac{50}{n^2} \frac{n(n+1)}{2} + \frac{25}{n^2} n - 4n \right]$$

use WolframAlpha simplify $((50/n^2)n(n+1)(2n+1)/6 - (50/n^2)n(n+1)/2 + (25/n^2)n - 4n)$

$$= \frac{5}{2n} \left(\frac{38n^2 + 25}{3n} \right)$$

$$A_n = \frac{190n^2 + 125}{6n^2}$$